Project Executable Files

SMARTBRIDGE - Project Executable Files

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Name:** AI FitTrack
- **Maximum Marks:** 3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)	Remarks / Details
1	**Complete source code** (all files and folders)	**Yes**	Full Node.js/Express.js backend source code included.
2	**README / Setup Guide** (instructions to run)	**Yes**	Complete README.md file located in the root directory.
3	**requirements.txt / package.json** / dependency file	**Yes**	package.json & package-lock.json included.
4	**Database schema / seed files** (if applicable)	**Yes**	Mongoose Schema files in src/models/User.js & src/models/Workout.js.
5	**Environment configuration file** (.env.example)	**Yes**	.env.example and configured .env included.

6	**Deployed application URL / Executable binary**	**Yes**	Local REST API server at http://localhost:5000.
7	**Dockerfile / Containerization config** (if applicable)	**NA**	Standard Node.js runtime environment used.
8	**APK** (if applicable)	**NA**	Backend RESTful API service.
9	**Test files and test results**	**Yes**	Postman Collection (FitTrack.postman_collection.json) & test logs.
10	**Demo video or walkthrough**	**Yes**	Google Drive walkthrough & API demo video links provided.

Step 2: File / Folder Structure

Plaintext

```
Code Files /
├── node_modules/
├── src/
│   ├── config/
│   │   └── db.js              # MongoDB connection configuration using Mongoose ODM
│   ├── controllers/
│   │   ├── aiController.js      # Handles Gemini AI recommendations and insights logic
│   │   ├── authController.js    # User registration, JWT login, and profile controllers
│   │   └── workoutController.js  # CRUD operations for workout management
│   ├── middleware/
│   │   ├── auth.js            # Express JWT authentication & token verification
│   │   └── errorHandler.js      # Centralized error response handling middleware
│   ├── models/
│   │   ├── User.js            # Mongoose schema for User collection (with bcrypt)
│   │   └── Workout.js           # Mongoose schema for Workout collection
│   ├── routes/
│   │   ├── aiRoutes.js          # Express router endpoints for AI services
│   │   ├── authRoutes.js        # Express router endpoints for authentication
│   │   ├── index.js           # Central route registry
```

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│   │   └── workoutRoutes.js      # Express router endpoints for workout CRUD
│   ├── services/
│   │   └── geminiService.js      # Google Gemini API (@google/genai) service handler
│   ├── app.js                    # Express application setup, CORS, and routing middleware
│   └── server.js                 # Server startup entry point and DB initialization
├── .env                          # Local environment variables (PORT, MONGO_URI,
GEMINI_API_KEY)
├── .env.example                  # Environment variable blueprint
├── .gitignore                    # Git ignore file for node_modules and .env
├── FitTrack.postman_collection.json # API collection for Thunder Client / Postman testing
├── package-lock.json             # Locked dependency versions
├── package.json                  # Project dependencies and npm startup scripts
└── README.md                     # Setup instructions, endpoints guide, and run commands
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	http://localhost:5000/api
Login Credentials (Demo)	**Email:** sathya@gmail.com **Password:** Sathya@01
Platform / Hosting Provider	Node.js Express Runtime & MongoDB Atlas Cloud Database
Repository Link	https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y

Demo Video Link	**Demo Video:** https://drive.google.com/file/d/1KswrctSaPI0eOYyJ3GkbbVHvHsU3EfZJ/view?usp=drive_link **API Testing Video:** https://drive.google.com/file/d/1TCXG48a-78ryxF3gipdRMwRBwuhAgPaa/view?usp=drive_link

Step 4: Run Instructions

Pre-requisites:

- **Node.js:** v16.0.0 or higher
- **npm:** v8.0.0 or higher
- **Database:** Active MongoDB Atlas Cloud Cluster or Local MongoDB
- **API Key:** Google Gemini API Key (@google/genai)

Execution Steps:

- **Step 1:** Open VS Code terminal and navigate into project directory:
- PowerShell

cd "Code Files"

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- **Step 2:** Install dependencies:
- PowerShell

npm install

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- **Step 3:** Configure environment variables in .env file:
- Code snippet

PORT=5000
MONGO_URI=mongodb+srv://kprince1720_db_user:Princekumar2005@cluster0.6160cws.mongodb.net/fittrack?retryWrites=true&w=majority
JWT_SECRET=super_secret_fittrack_jwt_key_12345
GEMINI_API_KEY=YOUR_GEMINI_API_KEY
GEMINI_MODEL=gemini-2.5-flash

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- **Step 4:** Start the REST API application server:
- PowerShell

npm start

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- **Step 5:** Confirm running server status in console:
- Plaintext

Server running in development mode on port 5000
MongoDB Connected: cluster0-shard-00-00.6160cws.mongodb.net

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- **Step 6:** Open Thunder Client or Postman and test endpoints at http://localhost:5000.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation PDF	Workaround / Status PDF
1	**ENOTFOUND MongoDB Connection Delay:** Occurs if non-SRV connection strings are used or during internet drops.	**Resolved:** Upgraded to standard mongodb+srv:// URI format with Mongoose connection retry options.
2	**Port Occupied (EADDRINUSE on Port 5000):** Port 5000 remains occupied by an orphaned process.	**Resolved:** Executed npx kill-port 5000 before running npm start.
3	**Gemini AI Deprecated Model Error (NOT_FOUND):** Older model strings fail API calls.	**Resolved:** Updated model string to gemini-2.5-flash in .env and geminiService.js.